

5. Project Development Phase

Objective

The Project Development Phase focuses on transforming the collected data into meaningful visualizations and interactive dashboards using Tableau. During this phase, the dataset was cleaned, calculated fields were created, worksheets were designed, dashboards were developed, and a Tableau Story was created to present business insights effectively.

Step 1: Import the Dataset

- Open Tableau Public/Desktop.
- Connect to the Food Ordering Behavior Dataset (Excel/CSV).
- Verify that all 50,000 records and 32 fields are imported successfully.
- Check data types and field names.

Output: Dataset successfully connected to Tableau.

Step 2: Data Preparation

The dataset was prepared before visualization by:

- Checking for missing values.
- Removing duplicate records (if any).
- Renaming fields for better readability.
- Assigning correct data types (String, Number, Date).
- Verifying numerical measures and dimensions.

Output: Clean dataset ready for analysis.

Step 3: Create Calculated Fields

Several calculated fields were created to enhance analysis:

Calculated Field	Purpose
Age Group	Categorize customers into age groups.
Delivery Speed Bucket	Categorize order preparation time.
Peak Period Group	Identify peak and off-peak order periods.
Rating Tier	Classify customer ratings.
Discount Flag	Identify discounted orders.
Order Value Category	Categorize orders based on value.
Total Revenue	Calculate total revenue generated.
Repeat Order Flag	Identify repeat customers.
Dimension Switch	Dynamic dashboard analysis.
Order Prep Time (Minutes)	Standardized preparation time.

Output: Calculated fields successfully created.

Step 4: Develop Worksheets

The following worksheets were created to analyze different aspects of the dataset:

Worksheet	Purpose
City Revenue	Revenue generated by each city.
Cuisine Demand	Popular cuisines based on order count.
Peak Meal Times	Order distribution across meal periods.
Order Value by Meal Type	Compare order value among meal types.
Order Prep Time Dist.	Distribution of order preparation times.
City Workload Heatmap	City-wise order workload analysis.
Rating vs Prep Time	Relationship between ratings and preparation time.
Repeat Orders	Analysis of repeat customer orders.
Rainy Weather Impact	Weather impact on ordering behavior.
Discount Impact	Effect of discounts on order volume.

Output: Interactive worksheets created successfully.

Step 5: Create KPIs

Key Performance Indicators (KPIs) were created to summarize business performance:

- Total Orders, Total Revenue, Average Order Value, Average Rating, Average Order Preparation Time, Repeat Order Percentage, Peak Period Orders, Slow Order Percentage.

Output: KPI cards added for dashboard summary.

Step 6: Design Dashboards

Four interactive dashboards were developed:

- Dashboard 1 (Business Performance): Total Revenue, Total Orders, Average Order Value, City Revenue, Cuisine Demand, Peak Meal Times.
- Dashboard 2 (Operational Efficiency): Order Prep Time Distribution, Peak Period Analysis, City Workload Heatmap, Rainy Weather Impact.
- Dashboard 3 (Customer Experience): Customer Ratings, Repeat Orders, Discount Impact, Cuisine Preference, Rating vs Prep Time.
- Dashboard 4 (Executive Summary): Overall KPIs, Business Insights, Dashboard Navigation, Summary Charts.

Output: Interactive dashboards completed successfully.

Step 7: Add Dashboard Interactivity

Interactive features implemented: Global Filters, Dashboard Filter Actions, Highlight Actions, Dynamic Parameters, Tooltips, Legends, Dashboard Navigation Buttons.

Output: Dashboards became interactive and user-friendly.

Step 8: Create Tableau Story

A Tableau Story was developed to present insights sequentially.

Story Points: Business Performance, Customer Ordering Trends, Peak Demand Analysis, Operational Efficiency, Customer Experience, Business Recommendations.

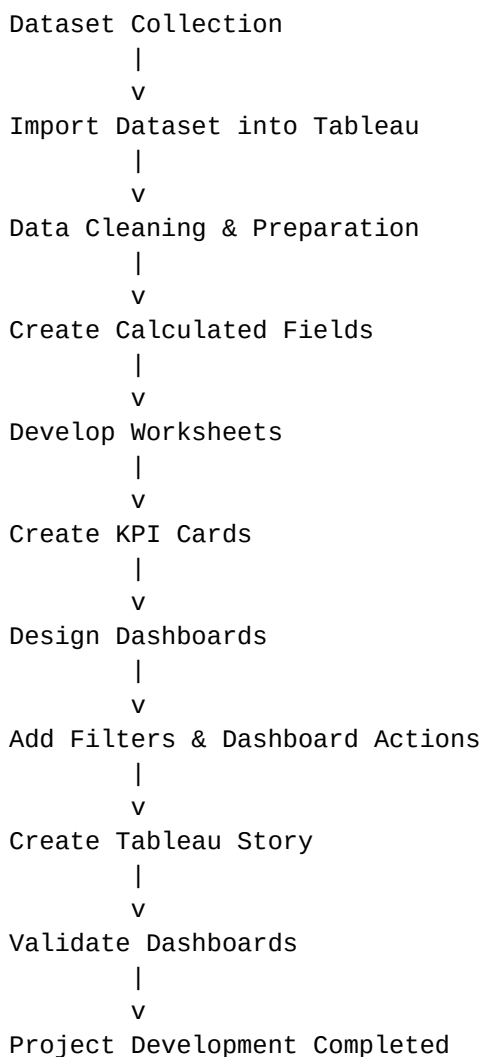
Output: Interactive Tableau Story completed.

Step 9: Validate the Dashboard

The dashboards were tested to ensure charts displayed correct info, filters updated all visualizations, and calculated fields produced accurate results.

Output: Dashboard validation completed successfully.

Project Development Workflow Diagram



Development Summary

The Project Development Phase involved importing and preparing the Food Ordering Behavior Dataset, creating calculated fields, developing interactive worksheets, designing KPI cards, building four dashboards, and creating a Tableau Story. Interactive filters, actions, and parameters were implemented to improve usability. The completed Tableau solution provides clear business insights into customer ordering behavior, operational efficiency, and overall

